

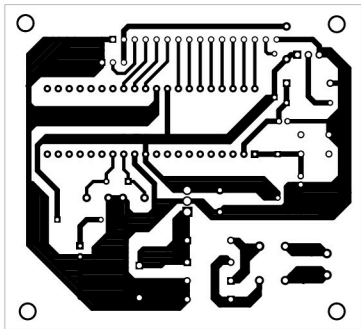


Fig. 2: Actual-size PCB pattern of the temperature indicator

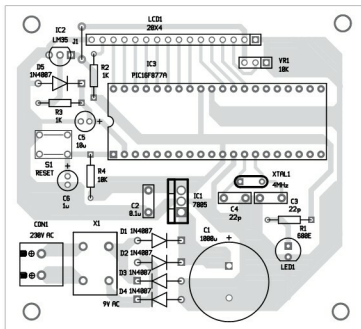


Fig. 3: Component layout of the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

controller (MCU) PIC16F877A (IC3), a 20 × 4 LCD display module (LCD1) and a few discrete components. LM35 is a precision integrated circuit temperature sensor, whose output voltage is linearly proportional to Celsius (Centigrade) temperature.

This sensor has a linear +10.0 mV/°C scale factor. It operates from 4V to 30V. It measures from -55 to +150°C temperature range. LM35 output is given to the analogue-to-digital converter (ADC) of the MCU. This Centigrade temperature is converted to Fahrenheit (°F), Kelvin (°K) and Rankine (°Ra), and displayed on the LCD module.

Analogue channel AN0 of the MCU receives the analogue signal from the temperature sensor for conversion to its 10-bit digital equivalent. A 4MHz crystal is connected to pins 13 and 14 of the MCU to provide a basic clock frequency.

Active low-reset function is provided by combining resistor R4 and capacitor C6. Switch S1 is used for the manual resetting of the MCU.

230V AC mains voltage is stepped down by transformer X1 to deliver the secondary output of 9V, 500mA. Transformer output rectified by full-wave rectifier comprising diodes D1 through D4 is filtered by capacitor C1. DC output is regulated at +5V by IC1. Capacitor C2 is used to bypass ripples, if any, in positive supply.

Construction and testing

An actual-size, single-side PCB for the MCU based temperature controller is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on the PCB to minimise assembly time and errors. Use an IC base for the MCU.

Software

The program is written in C language and compiled using HI-TECH compiler along with MPLAB to generate a hex code. The generated hex code is burnt into the MCU using a programmer with configuration bits as shown in Fig. 4. The program is well commented and easy to understand.

The software program consists of lcd.h header file and many other functions. The header file is used to control the LCD module for initialising

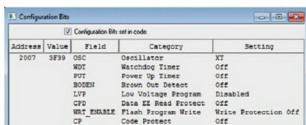


Fig. 4: Configuration bits

and to display the various messages and datum such as temperature. There are many functions like delay, hex-to-bcd, display, ADC_Init, ADC_Read and Temperature that are used in the program for various functions.

Delay function is used to provide the delay as per variable time. ADC of MCU PIC16F877A is initialised for ADC operation by ADC_Init function. Function ADC_Read takes the analogue signal, converts it into digital data and returns for analogue temperature equivalent digital data.

Converted digital data is converted into bcd format by hex-to-bcd function. Temperature function is used to convert ADC data into various temperature parameter units. Display function is used to display the data on the LCD. ●



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